

Yongye(Felix) Hu

☎ 510-277-7666 | ✉ hyy.felix@gmail.com | 🌐 felix-hyy | 👤 felix-hu.space

EDUCATION

University of California, Berkeley

Master of Engineering | Robotics Engineering

Aug 2024 - May 2025

Berkeley, CA

Wuhan University of Science and Technology

Bachelor of Engineering (Honours) | Mechatronics Engineering

Sep 2020 - May 2024

Wuhan, China

WORK EXPERIENCE

Wuhan Xiaoxian AI Technology Co., Ltd.

Mechatronics Engineer Intern

Sep 2023 - Mar 2024

Wuhan, China

- Developed control programs for a six-axis CRP2 robotic arm using embedded C and ladder logic, enabling automated multi-axis motion for fuel rod testing.
- Designed an STM32F103-based gateway to enable Modbus RTU communication between the PC and robotic controller, ensuring stable and real-time data exchange.
- Proposed and implemented a magnetically-triggered encoder-based imaging system to replace low-precision pulse triggering, ensuring ± 1 mm positional accuracy and reducing point-cloud data loss by 30%.

PROFESSIONAL EXPERIENCE

ROBOCON: Multi-Robot Coordination System

Integrated Mechanical Engineer

Sep 2022 - Jul 2023

Wuhan, China

- Developed embedded C firmware for STM32F407, enabling CAN-based motor control, BLE remote communication, and PID position control with ± 2 mm tracking error for an omnidirectional mobile robot.
- Implemented a custom UART protocol with headers, footers, and float unpacking; leveraged DMA + interrupts for non-blocking 100 Hz updates of 6-DoF data, with $< 0.2\%$ error rate.
- Developed and tuned a dual-mode chassis control architecture with three PID loops managing translation and orientation, achieving < 1.2 s error convergence and 40% improvement in directional stability.

Lightweight, Sustainable Remote-Mining Excavator System

R&D Engineer

Jan 2021 - Jul 2023

Wuhan, China

- Designed a compact, low-power crane system with ESP32-based Wi-Fi remote control, enabling low-latency communication and autonomous coordination across multiple devices, achieving over 95% task repeatability.
- Independently developed an STM32-based control system using fuzzy PID algorithms for closed-loop step-per motor control, achieving ± 1 mm positioning accuracy.
- Consolidated five functional boards into a single 4-layer PCB with KiCad, integrating control, power, and communication modules; reduced overall footprint by 40% and verified signal integrity via 100-MHz oscilloscope.
- Designed a multi-servo gripper system with ultrasonic radar for 360° object detection, capable of grasping up to 3 objects simultaneously.

L.I.F.T: A Hybrid Flying and Driving Robot Platform

Robotics Engineer & Project Status Manager

Sep 2024 - May 2025

Berkeley, CA

- Integrated OptiTrack motion capture system with robotic platform, achieving < 1 mm position accuracy and < 10 ms latency, supporting real-time position feedback for closed-loop control in indoor environments.
- Developed a ROS2-based robotic system integrating a simulator and MPC controller, enabling precise obstacle avoidance and full trajectory completion with real-time state logging.

SKILLS

Embedded Systems: STM32, ESP32, Arduino, Raspberry Pi, Embedded C/C++, PID/Fuzzy PID, Motor Control, PCB Design (KiCad)

Communication Protocols: CAN, UART, SPI, I2C, Modbus RTU, BLE, Wi-Fi, DMA

Programming: Python, C++, Linux, ROS2, MATLAB, Qt, LaTeX

Design & Prototyping: SolidWorks, Autodesk Fusion, Ansys, FDM/SLA 3D Printing, CNC Machining